Printing Guide

VERIFIED_DIGITALLY for part geometry, stored 3MF units, and bounding boxes. Actual print time, material use, shrinkage, warping, and airtightness are REQUIRES_PHYSICAL_VALIDATION.

Complete-part warning

You must print **both** the custom Ultra parts and the official Squircle upper stack. The six mandatory official files are in OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/; they are included in the release and listed in the table below.

Do **not** print the official original speaker chamber, original speaker plate, or original anti-slip ring. main_cabinet.3mf, the Ultra driver/radiator hardware, base_skirt.3mf, and anti_slip_ring.3mf replace those parts.

O07 and O08 are optional multi-material inserts. If using them, load O05, O07, and O08 together in the slicer and omit O06. For an ordinary single-material printer, use O05 plus O06 and ignore the optional folder.

Minimum printer volume

• Limiting part: outer_shell.3mf, exactly 192.0 x 212.0 x 189.0 mm in its required upright orientation.

• Absolute usable travel: 212 x 192 x 189 mm (X/Y may be swapped).

• Practical minimum: 220 x 220 x 200 mm only when the full 220 mm is usable; this leaves 4 mm per long-side edge and limits the shell brim to 3 mm.

• If purge lines, bed clips, firmware exclusions, or a wider shell brim reduce usable travel below 218 mm, use a larger printer. A 230 mm bed is preferable, but it is not the geometric minimum.

• In-plane rotation cannot fit the shell on a 210 x 210 mm bed. Side printing is unsupported because it creates extensive support contact and degrades slots and cosmetic surfaces.

Authoritative slicer baseline

• ASA primary; PETG alternative. Do not mix materials within a bolted joint.

• 0.4 mm nozzle; 0.20 mm layer; 0.45 mm line width.

• Five walls; six top and six bottom layers; 35% gyroid.

• ASA: 250-260 C nozzle, 100-110 C bed, enclosed printer, low part cooling, draft shield if chamber is below 40 C.

• PETG: 235-250 C nozzle, 75-85 C bed, moderate cooling after layer three.

• Supports: disabled. The horizontal acoustic openings are self-progressing circular overhangs; inspect their upper arcs and never place support on a gasket seat.

• Seam: rear (+Y) for cabinet/shell/shroud; away from all gasket lands.

• Elephant-foot compensation: set in the slicer from your coupon, not by sanding the part.

Print order and exact orientation

Group	Filename	Qty	Material	Face/orientation	Supports	Brim	Difficulty	Calibration
cosmetic	anti_slip_ring.3mf	1	TPU 95A	flat	none	none	2/5	yes

Group	Filename	Qty	Material	Face/orientation	Supports	Brim	Difficulty	Calibration
cosmetic	outer_shell.3mf	1	ASA	upright, base band on the bed	none	3 mm maximum on a 220 mm bed	5/5	yes
structural	main_cabinet.3mf	1	ASA	upright, acoustic floor on bed	none	10 mm	4/5	yes
structural	pressure_divider.3mf	1	ASA	flat, acoustic face on bed	none	none	2/5	yes
cosmetic	electronics_shroud.3mf	1	ASA	wide divider end on bed	none	none	2/5	yes
structural	active_driver_clamp_ring.3mf	1	ASA	lip face on bed	none	none	2/5	yes
structural	passive_radiator_clamp_ring.3mf	2	ASA	lip face on bed	none	none	2/5	yes
calibration	cable_gland.3mf	1	TPU 95A	body end on bed	none	5 mm	2/5	first
service tool	leak_test_adapter.3mf	1	TPU 95A	flange on bed; temporary service tool, not installed in service	none	none	2/5	yes
structural	base_skirt.3mf	1	ASA	service opening on bed	none	none	2/5	yes
structural	bottom_service_plate.3mf	1	ASA	exterior face on bed	none	none	2/5	yes
structural	ballast_cartridge.3mf	1	ASA	tray floor on bed	none	none	2/5	yes
structural	ballast_cartridge_lid.3mf	1	ASA	tongue face on bed	none	none	2/5	yes
calibration	coupon_official_interface.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_active_driver.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_passive_radiator.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_heat_set_insert.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_gasket_base.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_gasket_cap.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first
calibration	coupon_cable_passage.3mf	1	ASA	largest flat face on bed	none	5 mm	2/5	first

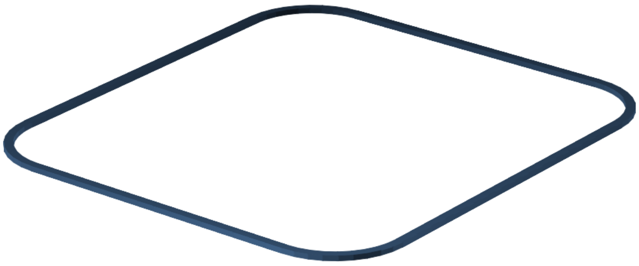
Group	Filename	Qty	Material	Face/orientation	Supports	Brim	Difficulty	Calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate_threads.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_pcb_spacer.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_lock_ring.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration
official required	OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate_snap_in_diffuser_ring.stl	1	ASA (PETG alternative)	lowest native-Z face on bed, as illustrated	none	none	2/5	after calibration

The 3MF files already store millimetres and the documented orientation.

CAD-derived orientation sheets

Print orientation — anti_slip_ring

BED = Z=0. flat. Keep sealing and insert faces free of support scars.

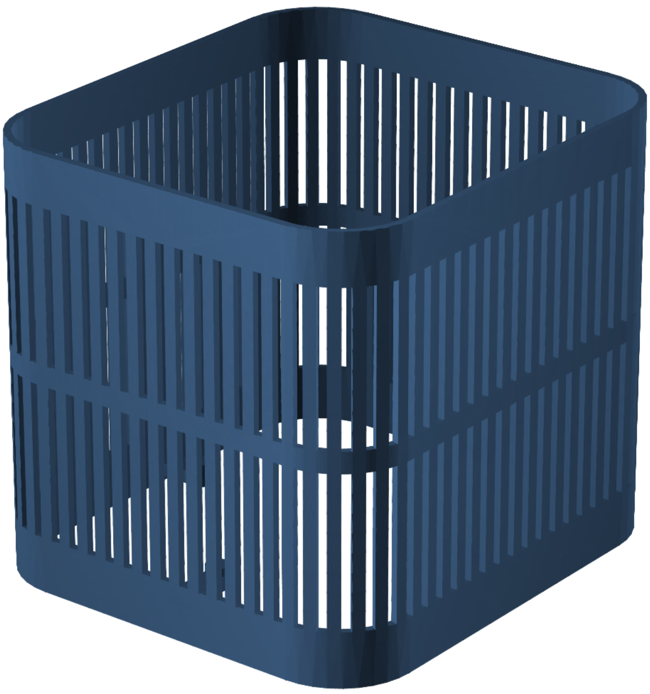


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

anti_slip_ring print orientation

Print orientation — outer_shell

BED = Z=0. upright, base band on the bed. Keep sealing and insert faces free of support scars.

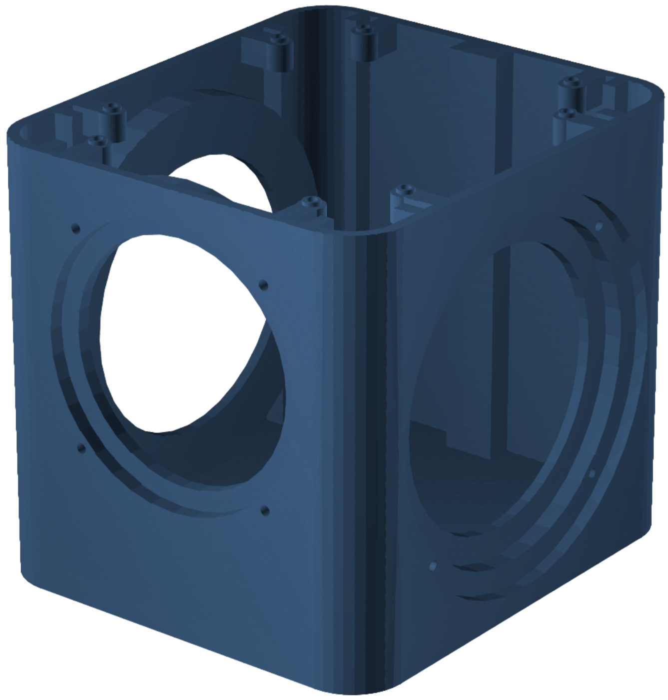


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

outer_shell print orientation

Print orientation — main_cabinet

BED = Z=0. upright, acoustic floor on bed. Keep sealing and insert faces free of support scars.

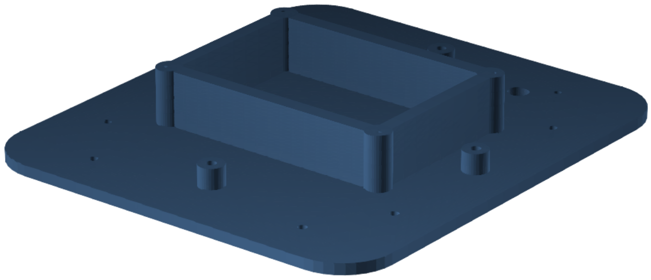


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

main_cabinet print orientation

Print orientation — pressure_divider

BED = Z=0. flat, acoustic face on bed. Keep sealing and insert faces free of support scars.

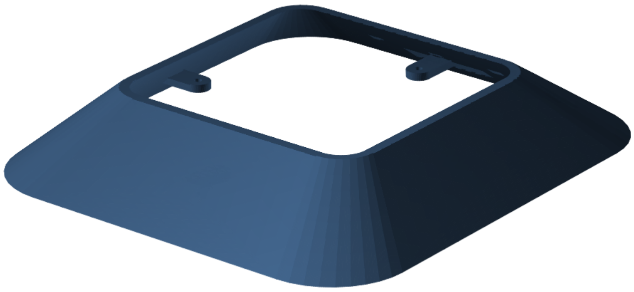


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

pressure_divider print orientation

Print orientation — electronics_shroud

BED = Z=0. wide divider end on bed. Keep sealing and insert faces free of support scars.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

electronics_shroud print orientation

Print orientation — active_driver_clamp_ring

BED = Z=0. lip face on bed. Keep sealing and insert faces free of support scars.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

active_driver_clamp_ring print orientation

Print orientation — passive_radiator_clamp_ring

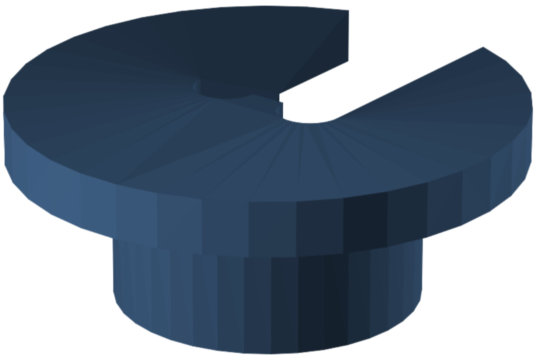
BED = Z=0. lip face on bed. Keep sealing and insert faces free of support scars.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

passive_radiator_clamp_ring print orientation

Print orientation — cable_gland

BED = Z=0. body end on bed. Keep sealing and insert faces free of support scars.

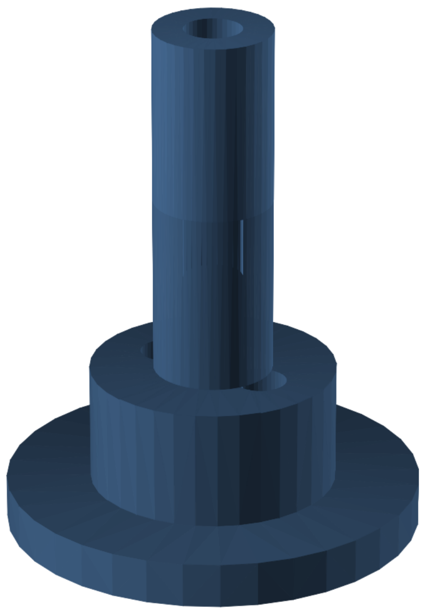


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

cable_gland print orientation

Print orientation — leak_test_adapter

BED = Z=0. flange on bed; temporary service tool, not installed in service. Keep sealing and insert faces free of support scars.

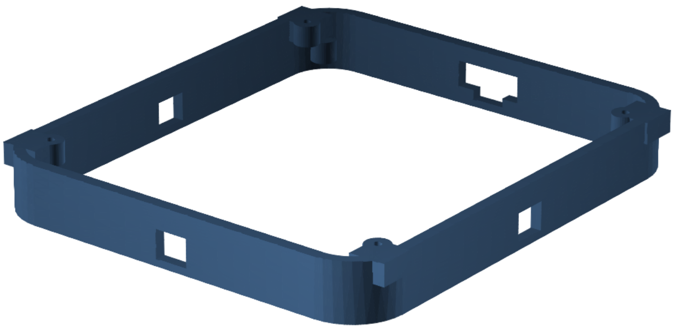


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

leak_test_adapter print orientation

Print orientation — base_skirt

BED = Z=0. service opening on bed. Keep sealing and insert faces free of support scars.

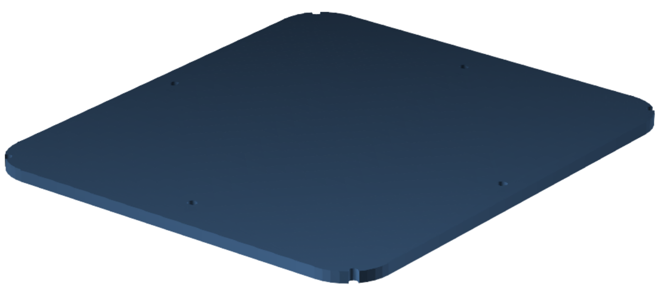


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

base_skirt print orientation

Print orientation — bottom_service_plate

BED = Z=0. exterior face on bed. Keep sealing and insert faces free of support scars.

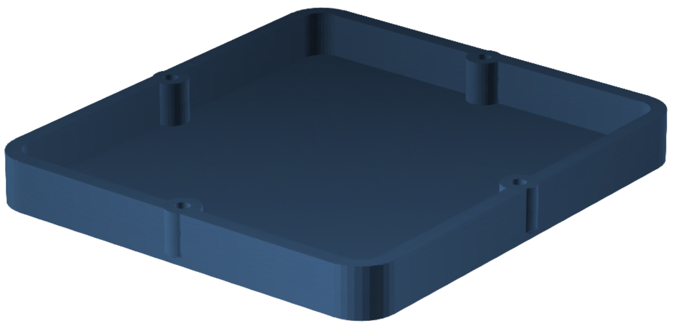


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

bottom_service_plate print orientation

Print orientation — ballast_cartridge

BED = Z=0. tray floor on bed. Keep sealing and insert faces free of support scars.

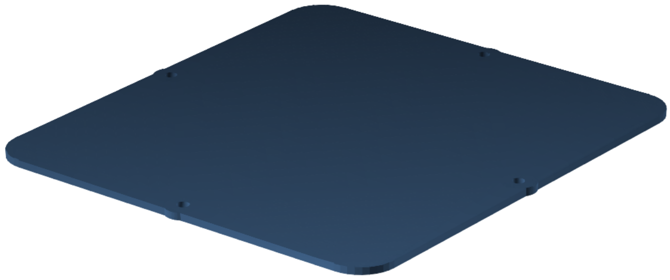


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

ballast_cartridge print orientation

Print orientation — ballast_cartridge_lid

BED = Z=0. tongue face on bed. Keep sealing and insert faces free of support scars.

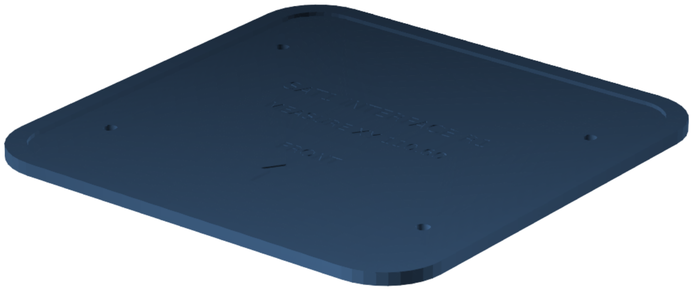


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

ballast_cartridge_lid print orientation

Print orientation — coupon_official_interface

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_official_interface print orientation

Print orientation — coupon_active_driver

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_active_driver print orientation

Print orientation — coupon_passive_radiator

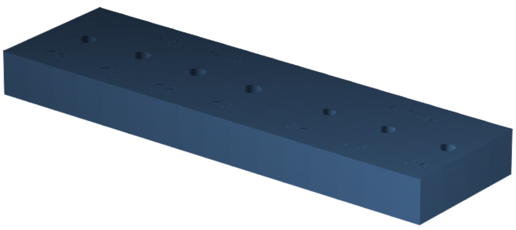
BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_passive_radiator print orientation

Print orientation — coupon_heat_set_insert

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.

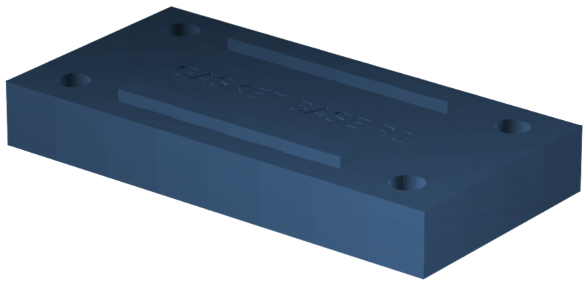


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_heat_set_insert print orientation

Print orientation — coupon_gasket_base

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.

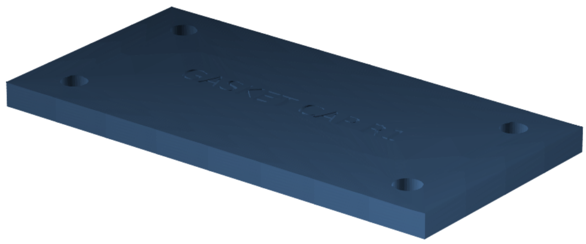


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_gasket_base print orientation

Print orientation — coupon_gasket_cap

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.

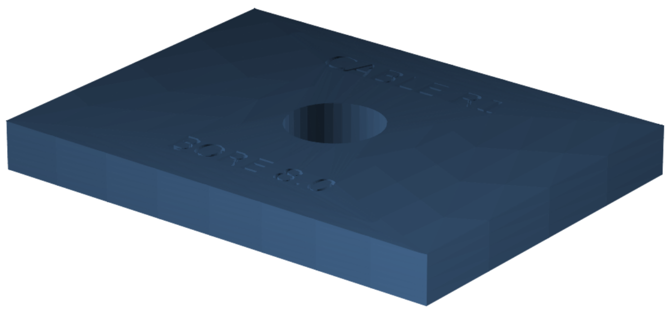


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_gasket_cap print orientation

Print orientation — coupon_cable_passage

BED = Z=0. largest flat face on bed. Keep sealing and insert faces free of support scars.



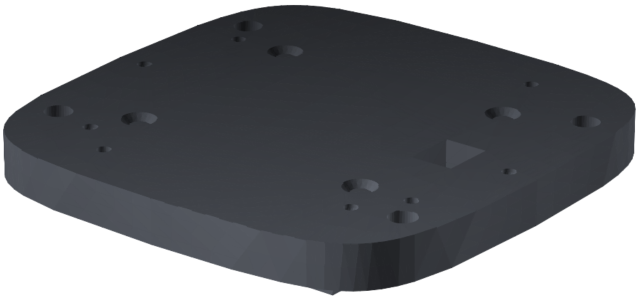
FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

coupon_cable_passage print orientation

Official Squircle orientation sheets

Official print orientation — official_mid_plate

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_mid_plate print orientation

Official print orientation — official_mid_plate_threads

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_mid_plate_threads print orientation

Official print orientation — official_pcb_spacer

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_pcb_spacer print orientation

Official print orientation — official_lock_ring

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_lock_ring print orientation

Official print orientation — official_top_plate

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_top_plate print orientation

Official print orientation — official_top_plate_snap_in_diffuser_ring

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_top_plate_snap_in_diffuser_ring print orientation

Official print orientation — official_top_plate_mm_buttons

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_top_plate_mm_buttons print orientation

Official print orientation — official_top_plate_mm_diffuser_ring

BED = lowest native-Z face, as shown. Preserve the official file unchanged; inspect all snap features and screw passages.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

official_top_plate_mm_diffuser_ring print orientation

Inspection before continuing

• Cabinet and divider: continuous, glossy-enough gasket lands; no seam gap, crack, under-extrusion, or insert bore opened into the chamber.

• Shell: all slots open, four retention bridges intact, no warp at either rim.

• Clamp rings: flat within 0.20 mm on a surface plate; lip clean and continuous.

• Ballast cartridge: lid holes align; both plates lie flat; four bosses sound.

• TPU parts: no tear, string in a wire bore, or layer split.

Approximate filament and time vary too much by slicer and machine to be verified digitally. Your slicer estimate is authoritative for planning; record it before starting each full-size print.

Source commit 6a7d13a7db83. No physical validation has been performed.